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University of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination - Semester I - UCSC AY22 [held in September/October 2025]

SCS 1307 – Probability and Statistics

(Two (2) Hours)

Answer ALL questions

Number of Pages = 9

Number of Questions = 4



To be completed by the candidate

Index Number:

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Important Instructions to candidates:

- I. Students should answer in the medium of English language only using the space provided in this question paper.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number CLEARLY on each and every page of this Question paper.
- IV. This paper consists of 4 questions in 9 pages (including the Cover Page).
- V. Answer ALL questions.
- VI. Programmable Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.
- VII. Non-Programmable calculators are allowed
- VIII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate

To be completed by the examiners

1	
2	
3	
4	
Total	

Question 1: This question consists of 10 multiple choice questions with five options with one correct answer. Underline the most suitable answer.

1. A fair die is rolled once. What is the probability of getting a number greater than 4?
 - a) $1/6$
 - b) $1/3$
 - c) $1/2$
 - d) $2/3$
 - e) $5/6$
2. What is the probability of selecting a vowel from the word 'PROBABILITY'?
 - a) $4/11$
 - b) $3/11$
 - c) $5/11$
 - d) $6/11$
 - e) $7/11$
3. A random variable X takes values 1,2, and 3. Probability of X takes the value 1 is 0.2 and the expected value is 2.1, what is the probability that X takes the value 2?
 - a) 0.5
 - b) 0.3
 - c) 0.1
 - d) 1.0
 - e) 0.33
4. Which of the following is true about mutually exclusive events?
 - a) They are always independent
 - b) $P(A \cap B) = 0$
 - c) $P(A \cup B) = 0$
 - d) $P(A|B) = P(A)$
 - e) $P(A \cap B) = P(A) * P(B)$
5. If a bag contains 3 red and 2 blue balls, and one ball is drawn at random, what is the probability that it is red given that it is not blue?
 - a) 0.2
 - b) 0.3
 - c) 0.5
 - d) 0.6
 - e) 1.0
6. If X is a random variable with values {1,2,3} and probabilities {0.2, 0.5, 0.3} . what is $E(X)$?
 - a) 1.5
 - b) 2.0
 - c) 2.1
 - d) 2.5
 - e) 3.0

7. What is the expected value of a fair four-sided (numbered 1 to 4) die roll?
- a) 3.5
 - b) 4
 - c) 2
 - d) 3
 - e) 2.5
8. A hospital has two types of wards: General and Emergency. Out of all in-house patients, 70% are in the General wards and 30% are in the Emergency wards. The probability of a patient needing surgery is 10% in the General ward and 40% in the Emergency ward. What is the probability that a randomly selected patient needs surgery?
- a) 0.12
 - b) 0.16
 - c) 0.19
 - d) 0.22
 - e) 0.25
9. A continuous random variable X follows the normal distribution with a mean of 4 and variance of 6. What is the probability that X takes the value 3?
- a) 0.05
 - b) 0.10
 - c) 0.25
 - d) 0.50
 - e) 0
10. If $P(A)=0.7$ and $P(B|A)=0.5$, what is $P(A \cap B)$?
- a) 0.5
 - b) 0.7
 - c) 1.2
 - d) 0.2
 - e) 0.35

(25 Marks)

Question 2: This question consists of 10 multiple choice questions with five answers. Underline the most suitable answer.

1. Which of the following scenarios best fits a binomial distribution?
- a) Measuring the height of students
 - b) Rolling a die and recording the number
 - c) Counting how many times a light bulb fails in a day
 - d) Surveying 100 people to see if they prefer tea or coffee
 - e) Recording temperature every hour
2. What is the standard deviation of a Poisson distribution with $\lambda = 4$?
- a) 2
 - b) 4
 - c) 8
 - d) 16
 - e) 1

3. Which of the following is NOT a characteristic of binomial distribution?
- a) Fixed number of trials
 - b) Two possible outcomes
 - c) Changing probability of success
 - d) Independent trials
 - e) Discrete outcomes
4. A student takes a 5-question multiple-choice quiz, with each question having 4 options and only one correct answer. What is the probability that the student guesses exactly 2 answers correctly?
- a) 0.263
 - b) 0.0878
 - c) 0.3125
 - d) 0.421
 - e) 0.5
5. A binomial random variable has $n=10$ and $p=0.4$. What are the mean and variance?
- a) Mean=4 and variance=1.5
 - b) Mean=4 and variance=2.4
 - c) Mean=6 and variance=2.4
 - d) Mean=6 and variance=1.5
 - e) Mean=4 and variance=3.6
6. What does the Central Limit Theorem (CLT) state?
- a) The population mean is always normally distributed
 - b) The sample mean of any distribution becomes normal as the sample size increases
 - c) The sample mean is equal to the population mean for all sample sizes
 - d) The population variance decreases with larger samples
 - e) The sample mean is always skewed
7. In a geometric distribution with probability of success $p=0.2$, what is the probability that the first success occurs on the third trial?
- a) 0.032
 - b) 0.064
 - c) 0.128
 - d) 0.256
 - e) 0.512
8. In a normally distributed dataset of student scores with a mean of 70 and a standard deviation of 10, approximately what percentage of students scored between 60 and 70?
- a) 68%
 - b) 95%
 - c) 99%
 - d) 34%
 - e) 47.5%

9. A population has a mean of 50 and a standard deviation of 16. What is the standard error of the sample mean for samples of size 32?
- a) 2.828
 - b) 2
 - c) 8
 - d) 0.707
 - e) 0.5
10. A university tracks the average GPA of randomly selected groups of 50 students. The population GPA distribution is not normal. According to the CLT, what can be said about the distribution of the sample means?
- a) It will be normal only if the population is normal
 - b) It will be approximately normal due to the large sample size
 - c) It will be skewed like the population
 - d) It will be normal
 - e) It cannot be determined

(25 Marks)

Question 3:

At SweetCrust Bakery, loaves of bread are inspected for the presence of air pockets that affect texture. The number of air pockets per loaf is known to follow a Poisson distribution with a mean of 0.8.

- (a) What is the probability that a randomly selected loaf has at least one air pocket?

(5 Marks)

- (b) A random sample of 10 loaves is taken from each of 20 batches. If more than 6 loaves in any batch have at least one air pocket, the entire batch is discarded. What is the probability that exactly 4 out of the 20 batches are discarded?

(10 Marks)

(c) At LuckyRoll Arcade, a player pays an amount (Rs. Y) and rolls two fair six-sided dice.

The payout rules are:

If the total is 7, the player wins Rs 10.00

If the total is 11, the player wins Rs 5.00

If the total is 2, the player wins Rs 3.00

For any other value player does not get anything.

To make the game fair, what amount should be paid by the player to start the game?

(10 Marks)

Question 4:

- (a) If a large number of samples of size n are taken from $\text{Poisson}(2.9)$. Approximately 1% of the sample means are greater than 3.41. Estimate n .

(9 Marks)

- (b) A certain type of pumpkin has a mass which is normally distributed with mean 1kg and standard deviation 0.15kg. In a lorry load of 800 of these pumpkins, estimate how many will have the mass greater than 1.13kg?

(8 Marks)

- (c) The masses of articles produced in a particular workshop are normally distributed with mean μ and variance σ^2 . 5% of the articles have a mass greater than 85g and 10% have a mass less than 25g. Find the values of μ and σ .

(8 Marks)

For negative values of z use $\Phi(z) = 1 - \Phi(-z)$

